SOURABH KUMAR DUBEY

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PROFESSIONAL SUMMARY

Computer Science undergraduate and competitive programmer with 700+ LeetCode problems solved (Knight badge) and a Codeforces peak rating of 1350+ (Pupil). Strong in Data Structures & Algorithms, with hands-on projects in network packet inspection, computer vision and full-stack AI tooling using Python, C++, PyTorch and JavaScript. Seeking software engineering internship opportunities.

EDUCATION

Bangalore Institute of Technology, Bengaluru — 2025 – Present
Bachelor of Engineering, Computer Science & Engineering | CGPA: 8.8

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript
- **Core CS:** Data Structures & Algorithms, Competitive Programming, Computer Networks, Problem Solving
- **Algorithms:** Graph Algorithms, Dynamic Programming (2D, Bitmask, DAG DP), Greedy, Backtracking, Sliding Window, Binary Search, Prefix Sums, Trees, Fenwick Tree, Segment Tree, SCC / Tarjan's Algorithm
- **AI / Vision:** PyTorch, YOLO, Ultralytics, Object Detection, Local LLMs (Ollama)
- **Web & Tools:** React, Vite, FastAPI, Three.js, Git, GitHub

PROJECTS

BuildFolio — AI-Powered Project Workspace | JavaScript, React, FastAPI, Ollama | github.com/skd-33/BuildFolio

- Built a workspace to build, track, understand and showcase technical projects, combining a project tracker with an AI portfolio generator.
- Developed a React/Vite frontend and FastAPI + SQLAlchemy backend, running a local LLM (Ollama, gemma3) for AI features; deployed a live demo on Render.
- Created for the DEV Hacktoberfest Weekend Challenge.

Deep Packet Inspection (DPI) System | C++ / Python | Computer Networks

- Built a Deep Packet Inspection system that analyzes network traffic at the packet level by parsing packet headers and traffic data.
- Iterated through multiple versions, applying core computer networking and cybersecurity concepts to inspect and classify network activity.

Smartphone-Based Obstacle Detection for Visually Impaired Users | Python, PyTorch, YOLO, Ultralytics

- Designed an obstacle detection application using a smartphone camera and YOLO (Ultralytics) with PyTorch for real-time object detection.
- Focused the system on delivering useful, timely obstacle information to visually impaired users.

Chess Square Guessing Game | HTML / JavaScript | github.com/skd-33/chess_square_guess_game

- Developed an interactive game for identifying chessboard squares, implementing game logic and user interaction with core data structures.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- **LeetCode:** Solved 700+ problems; earned the Knight badge (skd-33).
- **Codeforces:** Reached Pupil rank with a peak rating of 1350+ (sourabhkumardubey200).
- **Topics solved:** Graphs, dynamic programming (2D, bitmask, DAG DP), greedy, binary search, sliding window, prefix sums, trees, backtracking, combinatorics and number theory (sieve, inclusion–exclusion), optimizing for time and space complexity.
- **Advanced techniques applied:** SCC / Tarjan's Algorithm, Fenwick Trees, Segment Trees, Manacher's Algorithm, Booth's Algorithm.
- **Testing approach:** Validate solutions with brute-force stress testing before finalizing.
- **AWS Cloud Quest:** Placed 3rd in a tournament, one of 6 participants to complete it.
- **GitHub:** Earned Pull Shark and Quickdraw achievements.